

Lattices and Quadratic Forms

Lecture Notes

1 Even unimodular lattices

A lattice $L \subset R^n$ is *even* if $x \cdot x \in 2Z$ for all $x \in L$, and *unimodular* if $\det L = \pm 1$.

Theorem 1. *An even unimodular positive definite lattice of rank n exists if and only if $8 \mid n$.*

$$\theta_{E_8}(q) = 1 + 240 \sum_{n \geq 1} \sigma_3(n) q^n.$$

2 Discriminant forms

For an even lattice L the discriminant group is $A_L = L^\vee/L$ with the form $q_L(x) = x \cdot x \bmod 2Z$.